5 **(a)** State what is meant by the term *internal energy of an ideal gas*.

.....

..... [1]

(b) The pressure p exerted by an ideal gas is given by the equation

$$p = \frac{1}{3}\rho < c^2 >$$

State what the symbols ρ and $<c^2>$ represent.

ρ: .....

$<c^2>$: ..... [2]

For Examiner's Use

(c) Use the equation given in **(b)** to show that the total kinetic energy of the molecules of an ideal gas of volume V and pressure p is given by $\frac{3}{2}pV$.

[2]

(d) **(i)** Calculate the internal energy of an ideal gas of volume 3.4×10^{-4} m^3 when its pressure is 100 kPa.

internal energy = ... J [1]

(ii) Calculate the increase in internal energy when the gas expands at constant pressure to a volume of 8.8×10^{-4} m^3.

increase in internal energy = ... J [2]

(iii) Determine the heat supplied to the gas so as to enable the expansion at constant pressure to take place.

energy supplied = ... J [2]